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A hybrid additive with both suppressor and leveler capability for damascene copper electrodeposition

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ABSTRACT

Additives are essential for damascene copper electroplating to achieve defect-free deposition in nanoscale interconnect on chips. A hybrid additive named 1000EQ with capabilities of both suppressor and leveler has been synthesized and analyzed. The additive contains polyether segment and nitrogen cation which are typical functional groups of suppressor and leveler, respectively. Electrochemical measurements reveal that 1000EQ has strong inhibition toward copper deposition and is more resistant to the desorption of accelerator than traditional suppressor polyethylene glycol (PEG, Mw = 10,000). 1000EQ and bis-(sodium sulfopropyl)-disulfide (SPS) were used as additives to achieve a defect-free filling of features with 70 nm in width and an aspect ratio above 3. The ratio of copper film thickness over dense and sparse patterned area is close to 1 and the root mean square roughness (Rq) of the film is 9.4 nm, acting as a solid proof of its hybrid characteristic. The resistivity of the copper film is 1.80 ± 0.02 μΩ cm. The characterizations establish that 1000EQ is a promising candidate which can serve as both suppressor and leveler for damascene copper electrodeposition and is able to simplify the bath used in copper electrodeposition process.

1 Introduction

The interconnects with copper have been widely applied in the microelectronic industry for its better electromigration resistance and lower resistivity than aluminum [1–3]. Since the scale of interconnects with copper is shrinking as the miniaturization of electronic devices, the superfilling of nano-scale features such as vias and trenches without any defects like voids or seams remains a challenge [4]. Apart from achieving the defect-free superfilling (also so-called bottom-up filling) process, the surface planarization shares coordinate importance due to the uneven surface will cause difficulties to the chemical mechanical polishing (CMP) process.

Copper for interconnects is usually deposited by the additive-controlled electroplating. Three kinds of additives, categorized into accelerator, suppressor and leveler according to their different effects on

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copper deposition, are applied for accomplishing features superfilling and surface leveling during the electroplating process [5]. The superfilling of vias and trenches is fulfilled owing to the non-uniform distribution of accelerator and suppressor inside and outside the features [6]. Leveler is used for the planarization of the chip surface by avoiding the formation of bumps on dense field of vias and trenches by disturbing the continuous acceleration over the filled features. Difference between suppressor and leveler lies in their interplay with accelerator. Usually, suppressor is antagonistic to accelerator, whose inhibiting effectiveness can be counterbalanced by accelerator, while the inhibition of leveler can sustain in the presence of accelerator, and they can even behave synergistically in suppressing the deposition of copper [5].

Accelerator, which is mostly bis-(sodium sulfopropyl)-disulfide (SPS) and sodium 3-mercapto-1-propanesulfonate (MPS), can facilitate the copper plating process. It promotes the copper ion reduction [7–9] or replaces the inhibiting species on the plated copper surface, in other words, surface-adsorption coverage of different species determines the final electroplating rate. SPS can counterbalance the inhibitory effect of suppressors and enhance the deposition process by competitive adsorption on the electrode surface [10–12]. Suppressor, usually a polyether compound such as polyethylene glycol (PEG) and polypropylene glycol (PPG) or their copolymers, which are so-called polyalkylene glycol (PAGs) can diminish the copper electroplating rate. Both the molecular weight and structure have an impact on its inhibitory strength [13–17]. Some other compounds with ether chain were also developed as suppressor. Hai et al. combined imidazole and diglycidyl or PEG into one suppressor. The combination of the active functional groups such as ether, hydroxyl and imidazole led to a synergistic effect on suppressing the copper deposition process, finally achieving superfilling of submicron trenches together with SPS through the significantly improved suppression effect [18, 19].

Levelers generally have functional group like N^+ that allows the molecule to adsorb on bumps with dense negative charges, thereby decreasing the deposition rate of copper on bumps and finally realize leveling out the mounding sites on the surfaces. Commonly used levelers are Janus Green B (JGB) [20], Diazine Black (DB) [21], cationic surfactant such as *N*-hexadecyltrimethylammonium chloride (CTAC), dodecyltrimethylammonium bromide (DTAB) [22, 23] and polyquaternium-2 [24], and polycationic alkylamines like polyethylene imine (PEI) [25] which are applied for leveling out the undesired height over printed circuit board (PCB) or through holes (TH). New levelers serving as both suppressor and leveler at different applied current densities have been developed recently. Such levelers can significantly simplify the electroplating bath and cut down the organics used in the plating process. Hai et al. reported PVP as a switchable leveler additive for damascene applications. The properties of PVP could be manipulated by the electroplating current densities. It was similar to PAGs at lower current densities for implementing the superfilling process with SPS, while was more analogous to typical leveler at higher current densities for leveling process [26]. Likewise, Hai et al. also demonstrated IMEP as a hybrid suppressor combining abilities of both suppressor (type-I suppressor like PAGs) and leveler (type-II suppressor like PEI) at different applied current densities, successfully achieving the bottom-up filling of 130 nm wide trenches and surfaces planarization between areas with and without patterns [8, 27].

In this research, we combine the effective characteristics of traditional suppressor and leveler to one hybrid additive to achieve the superfilling and leveling process in a simpler way. The developed suppressor with polyether and N^+ functional groups has significant inhibition on copper deposition, besides, N^+ within its structure allows it to be more resistant to desorption of SPS, thereby having the leveling effect towards copper filling, finally resulting in the formation of defect-free features and planar surface between areas with diverse pattern densities. Such a combination not only realizes the efficient superfilling of vias and trenches and surface leveling by one additive but also lowers the complexity of the electrolyte bath, thereby simplifying the bath maintenance.

2 Experimental

2.1 Materials

The virgin make-up solution (VMS) containing 100 g/L $CuSO_4$, 10 g/L H_2SO_4, 50 ppm Cl^- and DI water was purchased from Shanghai Sinyang Semiconductor Materials Co., Ltd. (SYS2110). SPS and 4-chlorobutyryl chloride were purchased from Shanghai Aladdin Co., Ltd. PEG 1000, HCl and triethylamine were bought from Sinopharma Chemical Reagent Co., Ltd. PEG 10,000 was obtained from Energy Chemical. Ethyl acetate was bought from Shanghai Titan Scientific Co., Ltd. Bis(2-dimethylaminoethyl) ether was purchased from Shanghai Adamasi Reagents Co., Ltd. Dichloromethane was bought from Shanghai Dahe Chemicals Co., Ltd. All chemicals were used as received without any further purification.

2.2 Synthesis of 1000EQ

PEG 1000 (20 mmol) and triethylamine (40 mmol) were added into 50 mL dichloromethane in a three-neck round-bottom flask. 4-Chlorobutyryl chloride (40 mmol) dissolved in 10 mL dichloromethane was dripped into the former solution at a speed of 1 mL/min through a constant-pressure funnel under stirring with temperature at 0–10 °C during this process. Then the resulting solution continued to react for 2 h at room temperature. Following the removal of solvent by rotary evaporation, the resulting mixture was dissolved in ethyl acetate and washed with deionized water for several times. The organic phase was dried with $MgSO_4$. After filtration, ethyl acetate was removed by rotary evaporation to give the target intermediate as a light yellow liquid. ^{1}H NMR (400 MHz, DMSO-d_6, δ): 4.14 (t, 2 H), 3.66 (t, 2 H), 3.60 (t, 2 H), 3.51 (s, 36 H), 2.46 (t, 2 H), 1.97 (m, 2 H) (Fig. S1). The target intermediate was mixed with bis(2-dimethylaminoethyl) ether (20 mmol) directly in a round-bottom flask and the mixture was kept at 120 °C for 12 h. After reaction, 1000EQ was acquired as a light yellow solid. ^{1}H NMR (400 MHz, DMSO-d_6, δ): 4.15 (m, 2 H), 3.90 (m, 2 H), 3.61(m, 4 H), 3.51 (s, 36 H), 3.36 (m, 2 H), 3.10 (s, 6 H), 2.44 (t, 2 H), 1.93 (m, 2 H) (Fig. S2). According to NMR spectrum external standard method, the purity of the target material was about 90% (Fig. S3).

2.3 Electrochemical analysis

Electrochemical measurements including galvanostatic measurements (GMs) and cyclic voltammetry (CV) were performed in a three-electrode cell using an electrochemical workstation (CHI660, Shanghai Chenhua Instrument Co., Ltd.) which was connected with a rotating disk electrode (RDE) set-up (PINE, USA). A commercial Pt disc was used as the working electrode (5 mm in diameter). The electrode was smoothed by polishing down to 0.05 µm grit level before every test. A saturated mercury sulfate electrode (SSE) and a Pt wire were used as the reference electrode and counter electrode, respectively. GMs were conducted at a current density of J = − 20 mA/cm^2 and the rotation speed was fixed at 200 rpm. The electrode potential swept from − 0.8 to 0.2 V (vs. SSE) at a scan rate of 0.02 V/s when performing the CV test. All the electrochemical measurements were conducted in a glassy vessel containing 50 mL electrolyte solution.

2.4 Copper electrodeposition

Electroplating was conducted using two kinds of substrates (Shanghai Huali Microelectronics Co., Ltd), one of which was a planar silicon chip coated with a 120 nm copper seed layer and the other was a chip patterned with trenches and vias with a diameter of 70 nm and a depth of 150 and 220 nm. The patterned chips were also pre-coated with a copper seed layer of about 10 nm inside the trenches and vias and 40 nm outside for conductive purpose. The size of the planar chip was 2 × 2 cm, and the patterned chip was about 3.2 × 2.5 cm. Plating was performed under the program of 6.5 mA/cm^2 for 60 s and 60 mA/cm^2 for 60 s, and hot entry was used during the plating process. The solution was agitated using a magnetic stirrer with a rotational speed of 350 rpm.

2.5 Characterization

Nuclear magnetic resonance (NMR) was acquired with a 400 MHz spectrometer (Mercury Plus 400 M, Varian, Inc.). Gel permeation chromatography (GPC, Agilent1260, Agilent Technologies, Inc.) was used to analyze the molecular weight of the substance. Ion beam cross-section polisher (IB-09010CP, Japan Electronics Corporation) was used for cleaved samples polishing of the aimed sites in the tested chips. After polishing, the samples were imaged with a Scanning Electron Microscope (SEM, JSM-6701F, Japan Electronics Corporation and Zeiss Sigma HD, ZEISS) at the cross section for investigating the thickness of the plated copper film or the deposition result of the trenches and vias. The roughness of electroplated copper films on non-patterned chips were tested using an Atomic Force Microscope (AFM, Agilent 5420, Agilent Technologies, Inc.). The plated specimens were annealed at 150 °C for 10 min at the atmosphere of H_2 (5%)/N_2 (95%) with a heating rate of 10 °C/min before the AFM measurements. A four-point probe meter (SB100A/2, Shanghai Qianfeng Electronic Instrument Co., Ltd.) was utilized for electrical resistivity measurement. The data of sheet resistance and thickness were collected from at least four different locations. The probes spacing is 2 mm and the tested specimens were square with side length of 2 cm, thereby a corrector factor of 0.931 was induced to receive an accurate electrical resistivity.

3 Results and discussion

3.1 Synthesis

Polyether substances like PEG or PPG are generally used as efficient suppressors and leveling reagents containing N^+ group are found to provide reliable adsorption on the cathode surface by electrostatic attraction. Therefore, a reasonable combination of these two functional characters may give the new substance a hybrid characteristic especially there are significant differences between ether-containing suppressor and N^+-containing leveler in interacting with accelerator at different current densities. The molecular structure of the target compound, 1000EQ, and its synthetic route are shown in Fig. 1. The ester intermediate, obtained from the reaction of PEG 1000 and 4-chlorobutyryl chloride, was used without further purification to synthesize 1000EQ. The molecular structures of the intermediate and target compound were confirmed by ^{1}H NMR. Besides, the actual number of ethylene units in PEG 1000 was analyzed to be about 20 from the hydrogen integration. The GPC measurement reveals that the weight-average molecular mass and number-average molecular mass of the product are nearly 27,000 (Mw) and 14,000 (Mn), respectively. The polydispersity (PD) is nearly 1.88.

3.2 Electrochemical characterization

GMs and CV tests were performed to characterize the electrochemical properties of the 1000EQ on copper deposition. As shown in Fig. 2a, the injection of 10 ppm 1000EQ leads to a potential decrease of 140 mV, and the more the 1000EQ injected, more negative the potential is. A total potential decrease is about 230 mV after the injection of 100 ppm 1000EQ. Besides, as shown in the CV tests results in Fig. 2b, the reduction current of Cu^{2+} decreases with the injection of 1000EQ, and the peak potential shifts to a more negative potential. Such GMs and CV measurements results indicate that 1000EQ is an effective suppressor which can significantly inhibit the deposition rate of copper.

Figure 3a displays the potential-time curve of 1000EQ and PEG (Mw = 10,000) and their interactions with SPS. Such a time-dependent injection model was first developed by Rohan et al. [28] for detecting the adsorption and interplay of PEG and SPS, and it is then widely applied for additives measurements by other researchers [29–31]. One hundred ppm 1000EQ and 25 ppm SPS were injected into the VMS sequentially. The potential increases by $\Delta V_{1\text{–}2}$ = – 230 mV after the injection of 1000EQ, indicating a strong polarization ability of 1000EQ on copper deposition. After SPS being injected into the electrolyte, a decrease by $\Delta V_{2\text{–}2}$ = 130 mV is observed. The desorption rate of 1000EQ by SPS is $\eta = |\Delta V_{2\text{–}2}/\Delta V_{1\text{–}2}|$ = 56.5% and is about 12% less than the same indicator in bath recording the interaction of SPS and PEG ($\eta = |\Delta V_{2\text{–}1}/\Delta V_{1\text{–}1}|$ = 68.4%) which is generally used as the suppressor in filling small features. By the way, as Cl^- is also an important additive in inhibiting the copper deposition process, 5 ppm Cl^- was added into the electrolyte containing PEG to compensate the extra Cl^- introduced by the 1000EQ. As the curve shown in Fig. 3a, the extra Cl^- has little effect on the electrochemical behavior of PEG. A possible interpretation for this result is that the original 50 ppm Cl^- in VMS is high enough for building the binding sites for PEG. The CV curves of 1000EQ and PEG

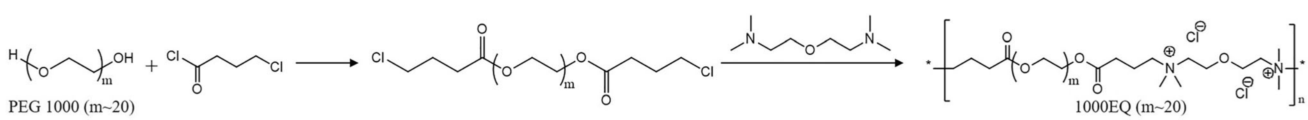

Fig. 1 The synthetic route of 1000EQ

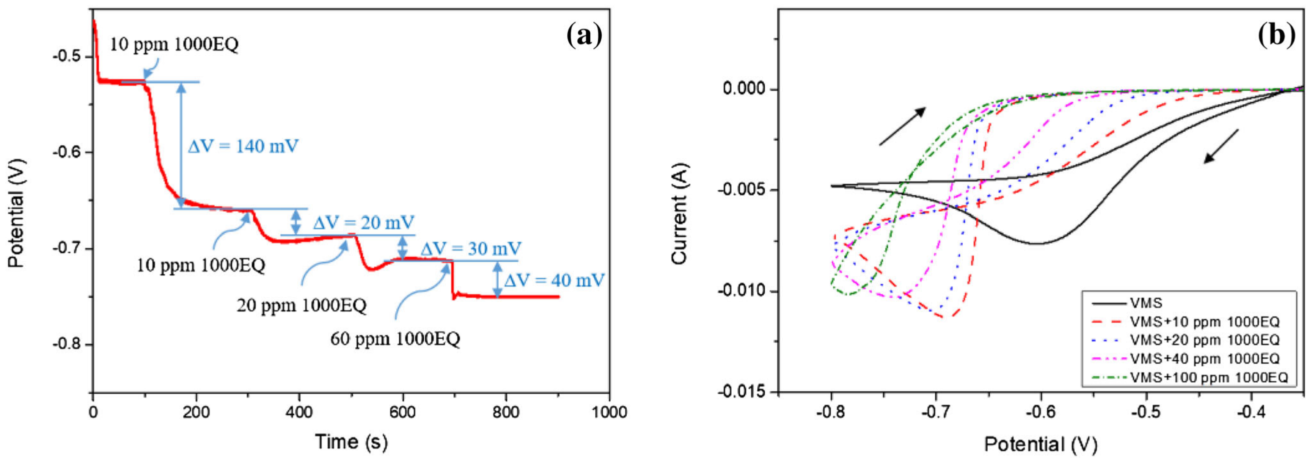

Fig. 2 Electrochemical results of 1000EQ in different concentrations: **a** GMs results; **b** CV curves

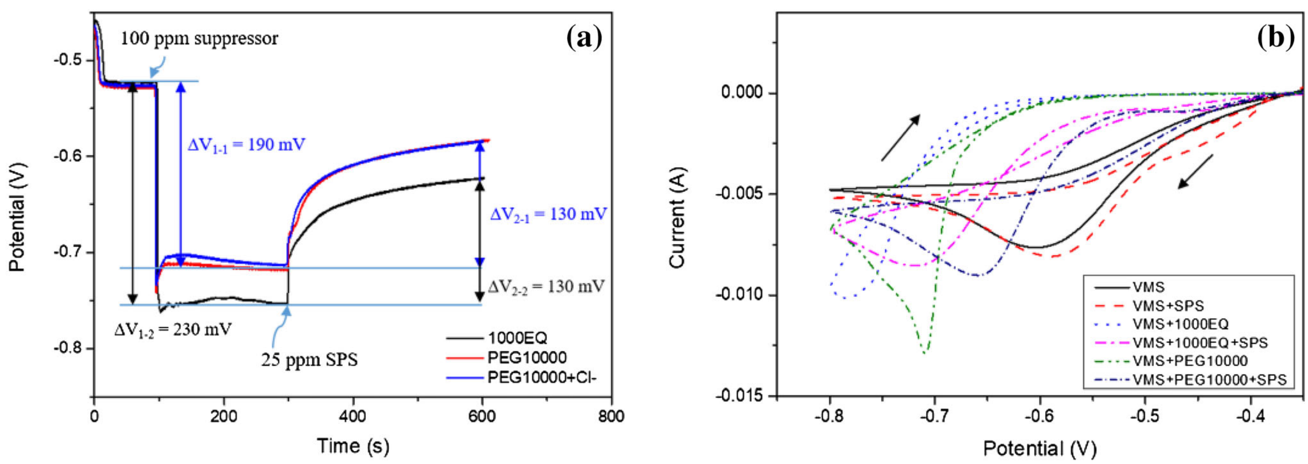

Fig. 3 Electrochemical results of the interplay of suppressor (1000EQ, PEG) and SPS: **a** GMs results; **b** CV curves (25 ppm SPS, 100 ppm 1000EQ/PEG 10000)

shown in Fig. 3b also indicate that 1000EQ is a stronger suppressor and more resistant to the competitive adsorption of SPS than PEG. Whether with SPS or not, the copper reduction current is smaller and peak potential is more negative in electrolyte with 1000EQ than that with PEG.

Furthermore, taking the actual case into consideration, the superfilling and leveling process are completed within a few minutes in damascene copper process and accelerator and suppressor are presented in the VMS at the same time when plating. Therefore, it will be better to measure the potential polarization under the circumstances of injecting SPS and 1000EQ simultaneously. As exhibited in Fig. 4, at the very beginning of the injection, a significant polarization is observed followed by a slow depolarization. Such an electrochemical behavior is resulted from a faster adsorption of 1000EQ and PEG on the electrode than SPS, while the following displacement by SPS results in the slow depolarization [28]. A steady potential increases by − 80 mV after the simultaneous addition of SPS and 1000EQ into the VMS. As a comparison, PEG has a lower potential increase of − 60 mV regardless of the absence or presence of the extra Cl^-. The result is consistent with the previously mentioned measurements in Fig. 3. It is well accepted that the adsorption of levelers comes from the interaction of N^+ within their structures with the negative-charge region on electrode [25, 32]. The positive-charged molecule is able to adsorb on the cathode by electrostatic attraction without being desorbed by SPS greatly, especially at higher current densities when there are more negative charges on the cathode.

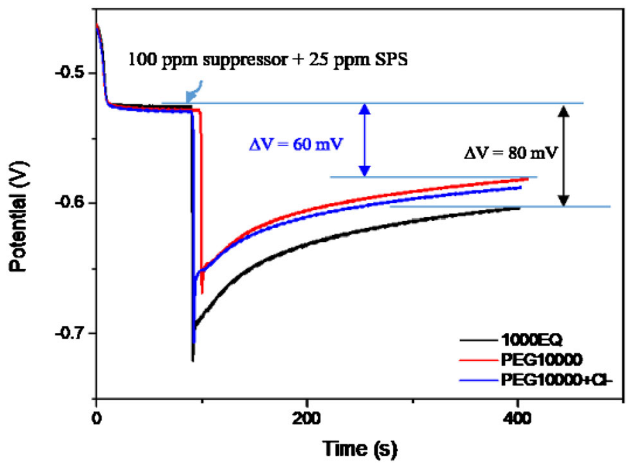

Fig. 4 GMs results of the interplay of suppressor (1000EQ, PEG) and SPS when the two kinds of additives were added into the electrolyte simultaneously

Hence, the improved adsorption and suppression ability of 1000EQ in competition with SPS is mostly resulted from the incorporation of N^+ group into the molecular structure.

3.3 Electroplating characterization

The superfilling and leveling capacities of 1000EQ were characterized by nano-scale features copper plating and results are presented in Fig. 5. Evident voids inside the 70 nm vias and trenches are observed after copper plating with 1000EQ-free VMS (Fig. 5a). The surface roughness of such a test specimen is measured to be 29.3 nm (Rq) (Fig. 5b). After the addition of 1000EQ, no defects, such as voids or seams are visible inside the features, indicating an outstanding capacity on superfilling of 1000EQ (Fig. 5c). Besides, the surface roughness of the deposited film is analyzed to be 10.2 nm (Rq) (Fig. 5d), certifying the successful leveling of surface and the remarkable leveling ability of 1000EQ. Thus, 1000EQ can function as a hybrid suppressor serving as a suppressor for superfilling process and a leveler for leveling step.

The leveling efficiency of 1000EQ (represented by an indicator of h_1/h_2) in patterned chips is 1.33, where h_1 and h_2 represent the copper film thickness over the dense patterned area with 70 nm wide features and sparse patterned field with larger-scale patterns, respectively. As a comparison, although PEG can also achieve the defect-free filling of 70 nm vias as shown in Fig. 6b, c, its leveling efficiency is 1.95. The improved leveling efficiency may come from the enhanced adsorption of 1000EQ on electrode surface from N^+ and confirms that 1000EQ has the hybrid capabilities of suppression and leveling.

Since organics may be embedded into the electroplated copper film, it is necessary to reduce the concentrations of organic additives in the bath. Hence, the superfilling and leveling capacities of 1000EQ at lower concentration were characterized as well. As shown in Fig. 7a–c, a combination of SPS (5 ppm) and 1000EQ (40 ppm, 50 ppm) leads to the formation of voids inside the nano-scale features, while the addition of 60 ppm 1000EQ and 5 ppm SPS results in a defect-free outcome. Figure 7d–f present the plated copper films over patterns and the leveling efficiencies of different concentrations combination of SPS and 1000EQ are displayed in Table 1. By means of reducing the ratio of SPS/1000EQ, the leveling efficiency is improved, and the bottom-up filling of 70 nm wide vias and trenches is finally achieved in a comparatively low concentrations of 5 ppm SPS and 60 ppm 1000EQ. To further characterize the leveling efficiency, the surface roughness was measured. As shown in Fig. 8, similar to the results received from the patterned chips, a good planarization is achieved even at lower concentrations of additives. It is clear that the synthesized 1000EQ enables the superfilling of small features and the leveling of chip surface successfully.

The resistivity of deposited copper film was also measured after annealing. The sheet resistance and thickness information of the electroplated non-patterned silicon chip are concluded in Table 2. The corresponding resistivity was calculated and displayed as well. No significant difference is observed in the four tested specimens and their resistivity stay low at a level of 1.80 μΩ cm or so. The resistivity is acceptably low compared to the resistivity of pure copper, which is 1.68 μΩ cm. The low resistivity of plated films indicates that the 1000EQ and SPS will not be greatly embedded into the deposited films and degrade the electrical properties. Therefore, it's validated that the synthesized 1000EQ is not only able to finish the bottom-up filling of nano-scale features and surface leveling, but also capable of achieving a good resistivity for conductive applications.

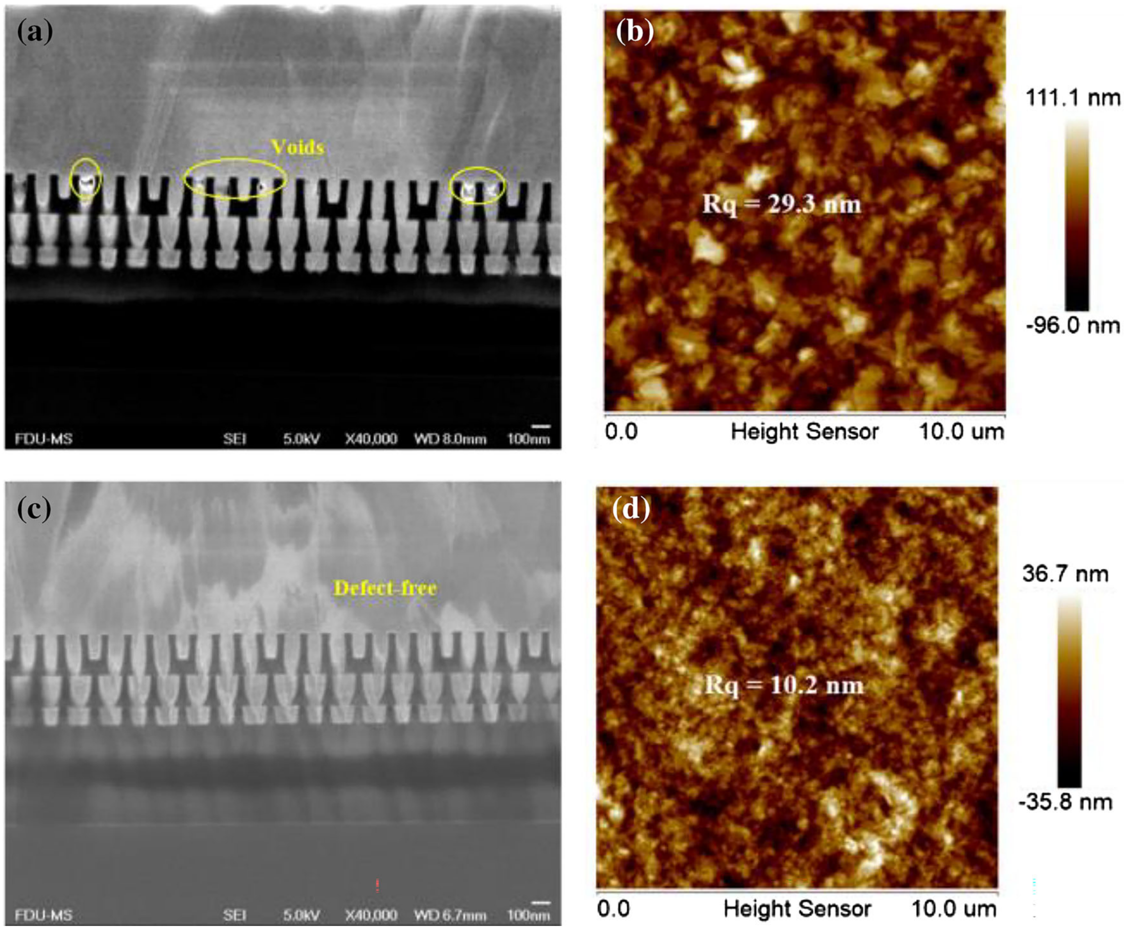

Fig. 5 Cross-sectional SEM images of 70 nm wide trenches and vias electroplated with copper from electrolyte containing: **a** VMS + 25 ppm SPS, **c** VMS + 25 ppm SPS + 100 ppm 1000EQ. AFM images of non-patterned Si chip electroplated with copper from electrolyte containing: **b** VMS + 25 ppm SPS, **d** VMS + 25 ppm SPS + 100 ppm 1000EQ

Fig. 6 Cross-sectional SEM images of 70 nm wide trenches and vias electroplated with copper from electrolyte containing: **a** VMS + 25 ppm SPS + 100 ppm 1000EQ, **b**, **c** VMS + 25 ppm SPS + 100 ppm PEG 10,000

4 Conclusion

A hybrid suppressor called 1000EQ, which can serve as both suppressor and leveler in copper electroplating, was successfully synthesized by combining both the efficient characteristics of suppressor and leveler. The electrochemical characterizations indicate that 1000EQ has stronger suppression effect and is more resistant to the competitive adsorption of SPS than PEG 10,000. Its stronger adsorption

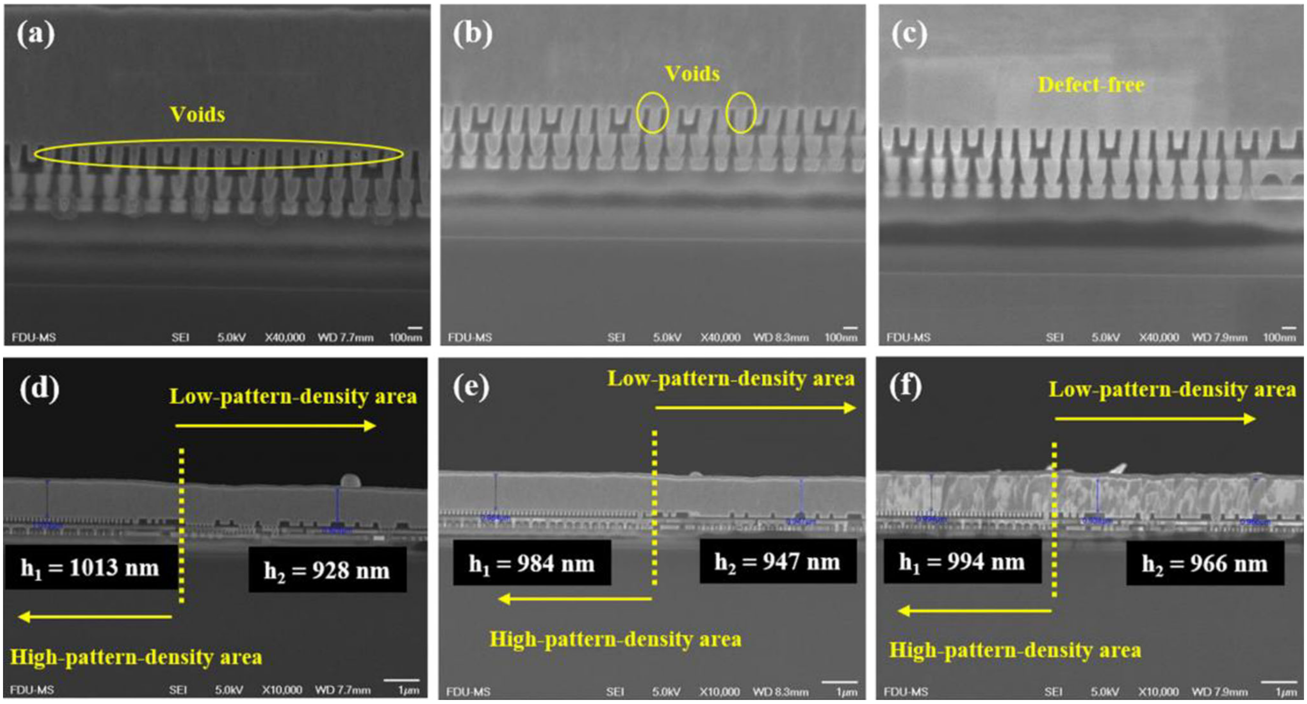

Fig. 7 Cross-sectional SEM images of 70 nm wide trenches and vias electroplated with copper from electrolyte containing: **a**, **d** VMS + 5 ppm SPS + 40 ppm 1000EQ, **b**, **e** VMS + 5 ppm SPS + 50 ppm 1000EQ, **c**, **f** VMS + 5 ppm SPS + 60 ppm 1000EQ

Table 1 Leveling efficiencies of chips deposited from electrolytes containing various concentrations of 1000EQ

Specimen NO.	SPS concentration (ppm)	1000EQ concentration (ppm)	Film thickness over high-pattern-density area (h_1, nm)	Film thickness over low-pattern-density area (h_2, nm)	Leveling efficiency (h_1/h_2)
1	5	40	1013	928	1.09
2	5	50	984	947	1.04
3	5	60	994	966	1.03

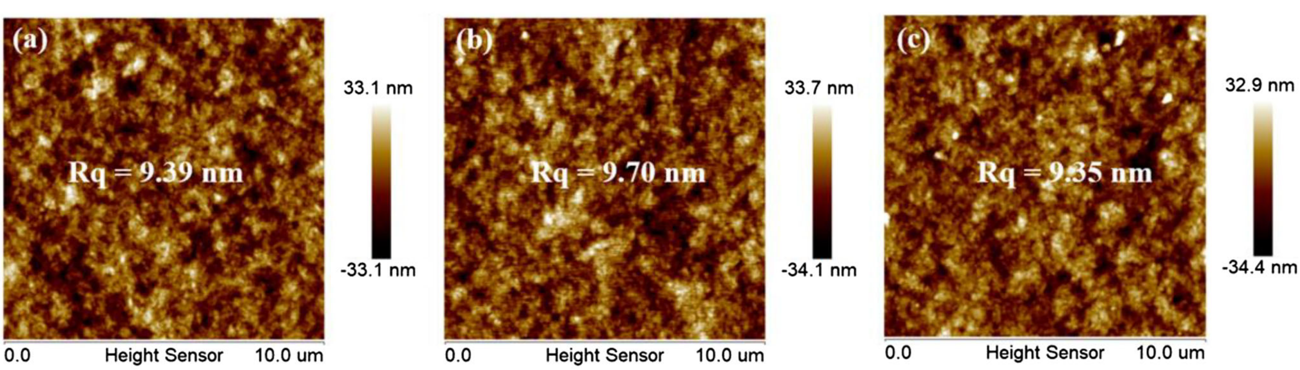

Fig. 8 AFM results of non-patterned chips electroplated with copper from electrolyte containing: **a** VMS + 5 ppm SPS + 40 ppm 1000EQ, **b** VMS + 5 ppm SPS + 50 ppm 1000EQ, **c** VMS + 5 ppm SPS + 60 ppm 1000EQ

Table 2 Sheet resistance and thickness of electroplated copper films from electrolyte containing different concentrations of 1000EQ and SPS

Specimen No.	SPS concentration (ppm)	1000EQ concentration (ppm)	Sheet resistance ($\Omega\ sq^{-1}$)	Thickness (μm)	Resistivity (μΩ cm)
1	25	100	0.0175	1.12	1.82 ± 0.03
2	5	40	0.0175	1.11	1.81 ± 0.01
3	5	50	0.0178	1.10	1.82 ± 0.02
4	5	60	0.0181	1.07	1.80 ± 0.02

characteristic possibly comes from the N^+. The electroplating experiments further verify such a hybrid character by accomplishing the bottom-up filling of 70 nm wide features with an aspect ratio up to 3 and greatly lowering the surface roughness of the plated copper film. An optimization in concentrations finally accomplished a good superfilling and leveling with 5 ppm SPS and 60 ppm 1000EQ with an acceptable resistivity of 1.80 ± 0.02 μΩ cm. Such a union of characteristic functional groups of suppressor and leveler into one single compound makes 1000EQ a hybrid suppressor and possible to simplify the electrodeposition bath. Thereby, it may lower containment embedded in the plated copper films, allowing a formation of thin films with relatively low resistivity, and will bring significant benefits for nano-scale interconnects. In summary, the synthesized 1000EQ is a promising candidate for copper deposition on state-of-art interconnects.

Author contributions

JC performed the experiments, analyzed data and wrote the manuscript. GL participated in synthesizing and characterizing the targeted 1000EQ. YZ participated in discussing some experiments data. YC and FX supervised the project and revised the manuscript. All authors have already read and approved the submitted manuscript.

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Data Availability

The datasets generated and/or analyzed during the current study are available from the corresponding author on reasonable request.

Declarations

Conflict of interest There is no conflict of interest to declare.

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